

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:20-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

20-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-22(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

For More Material Join: @JEEAdvanced_2024

PHYSICS

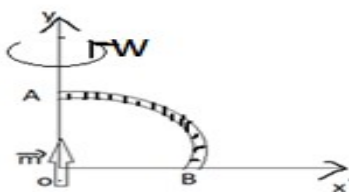
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

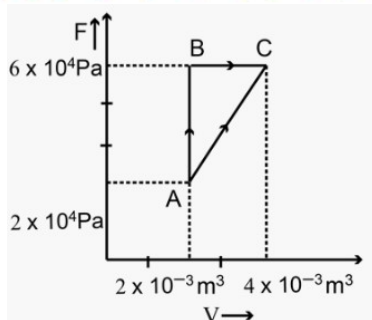
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In a Vernier calliper N divisions of Vernier scale coincides with $(N - 1)$ divisions of main scale (in which length of One division is 1mm) The least count of the instrument should be (in cm)
A) N B) $N-1$ C) $1/10 N$ D) $1/N-1$
2. An ideal gas is expanded adiabatically. The adiabatic exponent of gas $\gamma = \frac{4}{3}$. How many times the volume of gas needs to be increased to make root mean square speed of the gas half of its initial value?
A) 4 B) 2 C) 64 D) 16
3. Two identical lenses made of the same material of refractive index 1.5 have the focal length 12cm. These Lenses are kept in contact and immersed in a liquid of refractive index 1.35 The combination behaves as
A) converging lens of focal length 27 cm
B) converging lens of focal length 6 cm
C) converging lens of focal length 9 cm
D) converging lens of focal length 12 cm
4. Consider a quarter circular conducting ring of large radius r with its centre at the origin, where a magnetic dipole of moment $\vec{m}$ is placed as shown in the figure. If the ring rotates at constant angular velocity ω about the y -axis, electromotive force induced between its ends is



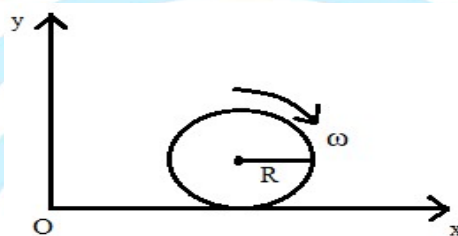
- A) Zero B) $\mu_0 m \omega / (2\pi r)$ C) $\mu_0 m \omega / (4\pi r)$ D) $\mu_0 m \omega / (8\pi r)$

5. The density of a cube is measured by measuring its mass and length of its sides. If the maximum error in the measurement of mass and length are 4% and 3% respectively, the maximum error in the measurement of density will be
- A) 7% B) 9% C) 12% D) 13%
6. With what velocity should a particle be projected from the surface of earth so that its height becomes equal to radius of earth
- A) $\left(\frac{GM}{R}\right)^{\frac{1}{2}}$ B) $\left(\frac{8GM}{R}\right)^{\frac{1}{2}}$ C) $\left(\frac{2GM}{R}\right)^{\frac{1}{2}}$ D) $\left(\frac{4GM}{R}\right)^{\frac{1}{2}}$
7. If a dipole of dipole moment $\hat{p}\hat{i}$ is placed at point $(0, y)$ and a point charge at origin of a coordinate system. Net electric field at point $(x, x+y)$ vanishes. If x and y both are positive, the coordinate y is equal to
- A) x B) $2x$ C) $2.5x$ D) $3x$
8. A series R-C circuit is connected to an alternating voltage source. Consider two situations:
- a) When capacitor is air filled.
- b) when capacitor is mica filled.
- Current through resistor is I and voltage across capacitor is V then:
- A) $V_a > V_b$ B) $i_a > i_b$ C) $V_a = V_b$ D) $V_a < V_b$
9. Figure below shows two paths that may be taken by a gas to go from a state A to a state C. In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be

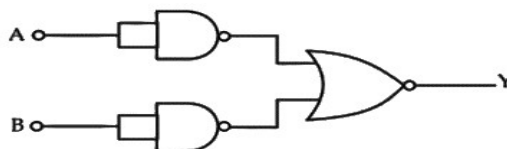


- A) 500 J B) 460 J C) 300 J D) 380 J

10. Two magnets of magnetic moments M and $2M$ are placed together in a vibration magnetometer with identical poles in the same direction. The time period of vibration is T_1 . If the magnets are placed with opposite poles together and vibrate with time period T_2 , then
- A) T_2 is infinite B) $T_2 = T_1$ C) $T_2 > T_1$ D) $T_2 < T_1$
11. In an electromagnetic wave in free space the root mean square value of the electric field is $E_{rms} = 6V/m$. The peak value of the magnetic field is:
- A) $2.83 \times 10^{-8} T$ B) $0.70 \times 10^{-8} T$ C) $4.23 \times 10^{-8} T$ D) $1.41 \times 10^{-8} T$
12. A spherical shell of 1 kg mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{a}{3} R^2 \omega$. The value of a will be

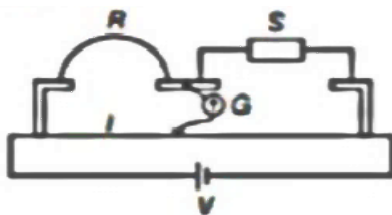


- A) 2 B) 3 C) 5 D) 4
13. A metallic surface is illuminated with radiation of wave length λ . The stopping potential is V_0 . If the same surface is illuminated with radiation of wave length 2λ . The stopping potential becomes $\frac{V_0}{6}$. The threshold wave length for this metallic surface will be.
- A) $\frac{\lambda}{4}$ B) $\frac{5\lambda}{2}$ C) λ D) $\frac{3\lambda}{2}$
14. Hydrogen atom is excited from ground state to another state with principle quantum number equal to 4. Then the number of spectral lines in the emission spectra will be.
- A) 2 B) 3 C) 5 D) 6
15. Identify the logic operation carried out by circuit as shown

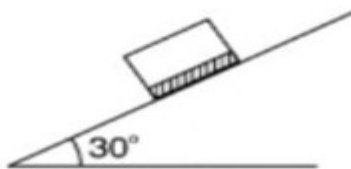


- A) OR B) AND C) NOR D) NAND

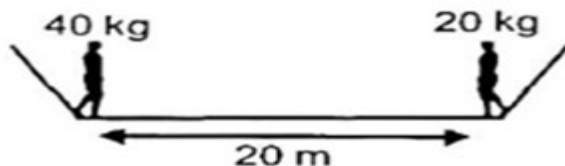
16. In a meter bridge experiment, S is a standard resistance and R is a resistance wire. It is found that, balancing length is $l = 25 \text{ cm}$. If R is replaced by a wire of half length and half diameter that of R of same material, then find the balancing distance l'



- A) 20 cm B) 30 cm C) 40 cm D) 50 cm
17. The internal radius of a 1 m long resonance tube is measured as 3.0 cm. A tuning fork of frequency 2000 Hz is used. The first resonating length is measured as 4.6 cm and the second resonating length is measured as 14.0 cm. Speed of sound in air is (m/s)
- A) 175 B) 276 C) 376 D) 425
18. A cubical block of side 2 m having mass 20 kg slides with constant velocity of 10 ms^{-1} on an inclined plane lubricated with the oil of viscosity $\eta = 10^{-1}$ poise. Taking $g = 10 \text{ ms}^{-2}$, calculate the thickness of layer of liquid



- A) 4 mm B) 2 cm C) 5 cm D) 6 mm
19. Two men of masses 40 kg and 20 kg are standing on a boat of mass 100 kg. Length of boat is 20 m. Neglect the friction between water and boat. Find the displacement of the boat when both person reach at middle of boat.



- A) $\frac{5}{4} \text{ m}$, towards right B) $\frac{5}{4} \text{ m}$, towards left
- C) $\frac{5}{8} \text{ m}$ D) $\frac{5}{8} \text{ m}$, towards left

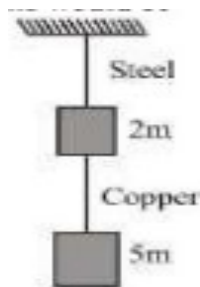
- ## SECTION-II

Marking scheme: +4 for correct answer, -1 in all other cases.

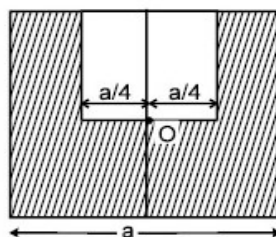
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- Page. No. 6

26. If the ratio of diameters, lengths and young's modulus of steel and copper wires shown in the figure are p, q and s respectively, then the corresponding ratio of increase in their lengths would be $\frac{Mq}{Nsp^2}$ then find value of $M + N$



27. In a diffraction pattern due to single slit of width 'a' the first minimum is observed at an angle 30° when light of wavelength $5000 \text{ \AA}$ is incident on the slit. The first secondary maximum is observed at an angle of $\sin^{-1}\left(\frac{x}{y}\right)$ then find the value of $x + y$ is :
28. A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to $0A$ from $4A$ in $0.05s$. If the resistance of the coil is $10\pi^2\Omega$. Find the total charge flowing through the coil during this time is (in μc)
29. A cannon shell moving along a straight line bursts into two parts. Just after the burst one part moves with momentum 40 Ns making an angle 30° with the original line of motion. The minimum momentum of the other part of shell just after the burst (in Ns)
30. A square plate of edge $a/2$ is cut out from a uniform square plate of edge 'a' as shown in figure. The mass of the remaining portion is M . The moment of inertia of the shaded portion about an axis passing through 'O' (centre of the square of side a) and perpendicular to plane of the plate is $\frac{pMa^2}{q}$ then find the value of $q - p$



SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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31. Among the following compounds the one which is stable is?

- A) FeI_3 B) CuI_2 C) TiF_2 D) $FeCl_3$

32. A complex is represented as $CoCl_3 \cdot xNH_3$. Its 0.1 m solution in aqueous solution shows $\Delta T_f = 0.558^\circ$ and assume 100% ionization and co-ordination number of Co(III) is six. What is the complex [$K_f(H_2O) = 1.86 \text{ mol}^{-1} \text{ K}$] ?

- A) $CoCl_3 \cdot 6NH_3$ B) $CoCl_3 \cdot 5NH_3$ C) $CoCl_3 \cdot 4NH_3$ D) $CoCl_3 \cdot 3NH_3$

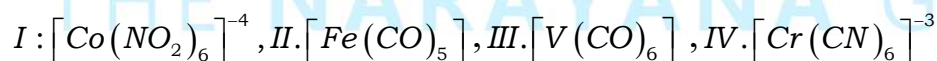
33. According to Bohr's atomic theory:

- 1) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$
- 2) The product of velocity (v) of electron and principal quantum number (n), ' vn ' $\propto Z^2$
- 3) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$
- 4) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$

Choose the most appropriate answer from the option given below:

- A) (3) only B) (1) only
C) (1) (3) and (4) only D) (1) and (4) only

34. Consider the following complexes:



Identify the set of complexes with one unpaired electron.

- A) I,II B) II , III C) III , IV D) I, III

35. Which of the following chemical reaction is not possible

- A) $F_2 + 2X^- \rightarrow 2F^- + X_2$ [$X = Cl, Br, I$] B) $Cl_2 + 2X^- \rightarrow 2Cl^- + X_2$ [$X = Br, I$]
C) $I_2 + 2X^- \rightarrow 2I^- + X_2$ [$X = F, Cl, Br$] D) $Br_2 + 2X^- \rightarrow 2Br^- + X_2$ [$X = I$]

A) Br_3O_8

C) Sodium thiosulphate

37. Select the INCORRECT statement regarding the complex $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$ is

A) The complex can't display optical isomerism.

B) It is a low spin complex

C) It has $t_{2g}^6 e_g^0$ type arrangement of electrons

D) Its magnetic moment is 0 BM

38. Column I (Element)

Column II (substance used for its estimation)

A) Carbon

p) conc. H_2SO_4

B) Sulphur

q) Fuming nitric acid

C) Nitrogen

r) Ammonium Molybdate

D) Phosphorus

s) Copper oxide

Choose the correct matching in the following

A) A - s, B - q, C - p, D - qr

B) A - q, B-s, C-p, D-q

C) A - p, B-q, C-s, D-p

D) A - q, B - p, C - s, D - q

39. Which one of the following statements is not correct?

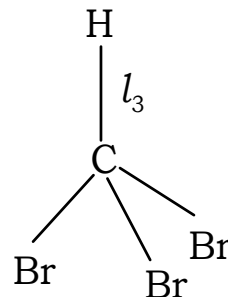
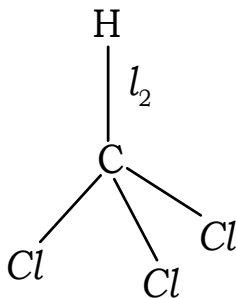
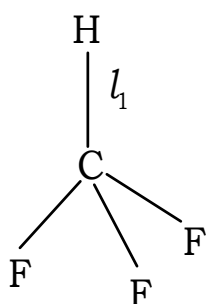
A) Energy of an electron in hydrogen atom depends only on principal quantum number

B) With increase in principal quantum number of an orbital energy of an electron in that orbital increases

C) Energy of an electron in 2s orbital of hydrogen atom is same as the energy of 2s orbital electron in Lithium atom.

D) In multi electron atoms energy of different subshells of a particular shell is different

40.

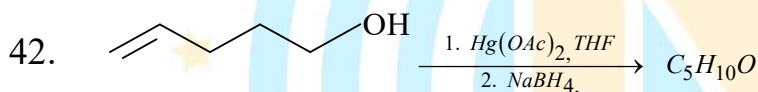


l_1 , l_2 and l_3 are the bond lengths of C – H bonds in the given molecules. Correct order of these bond lengths will be

- A) $l_1 > l_2 > l_3$ B) $l_3 > l_2 > l_1$ C) $l_1 > l_3 > l_2$ D) $l_1 = l_2 = l_3$

41. In a closed container solid NH_4HS was heated at 100°C . The following equilibrium is established with K_p for the equilibrium was found to be 25 atm^2 . $\text{NH}_4\text{HS}_{(s)} \rightleftharpoons \text{NH}_{3(g)} + \text{H}_2\text{S}_{(g)}$. The total pressure of the mixture of a gases at equilibrium is

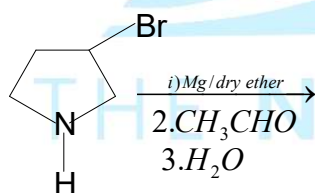
- A) 10 atm B) 5 atm C) 25 atm D) 625 atm



What is the most reasonable structure for the product of this reaction?

- A) B) C) D)

43. The final major product in the following sequence of reactions is



- A) B) C) D)

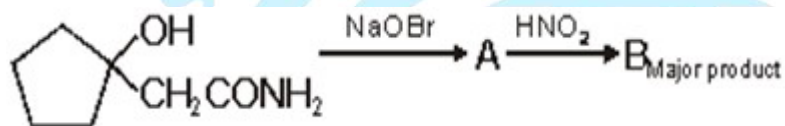
44. A compound 'A' with molecular formula $C_{10}H_{13}Cl$ gives a white ppt. on adding $AgNO_3$ solution. A on reaction with alcoholic KOH gives Compound B as main product. B on ozonolysis gives C and D. C gives Cannizzaro's reaction but not aldol condensation. D gives aldol condensation but not Cannizzaro's reaction. The formula of A is

- A) $C_6H_5CH_2-CH_2-CH_2-CH_2Cl$ B) $C_6H_5CH_2-CHCl-CH_3$
 C) $O-(CH_2Cl)C_6H_4(CH_2CH_2CH_3)$ D) $C_6H_5CH_2C(Cl)(CH_3)_2$

45. Myosin is the common example of

- A) Vitamin B) Fibrous proteins
 C) Carbohydrates D) Globular proteins

46. The major product formed in the following sequence of reactions

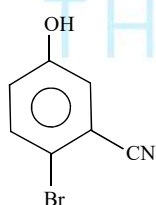


- A) B) C) D)

47. The enthalpy of combustion of hydrogen is $-249 \text{ KJ.mole}^{-1}$ and bond enthalpies of H-H, and O=O are 433 KJ.mole^{-1} and 492 KJ.mole^{-1} respectively. The bond enthalpy of O-H is

- A) $-282 \text{ KJ mole}^{-1}$ B) $-664 \text{ KJ mole}^{-1}$ C) 464 KJ mole^{-1} D) 232 KJ mole^{-1}

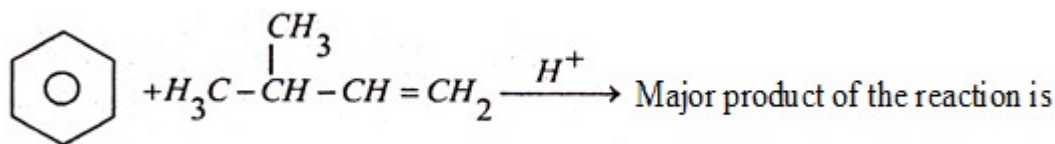
48. IUPAC name of the following compound is



- A) 4-bromo -3- cyanophenol B) 2-bromo-5-hydroxybenzonitrile
 C) 2-cyano-4-hydroxybromobenzene D) 6-bromo-3-hydroxybenzonitrile

49. The standard reduction potential for $\text{Cu}^{2+} / \text{Cu}$ is +0.34 V, calculate the reduction potential at $P^H = 14$ for the couple, K_{sp} of $\text{Cu}(\text{OH})_2$ is 1.0×10^{-19}
- A) 0.2205 V B) -0.325 V C) -0.2205 V D) 0.325 V

50.



- A)
- B)
- C)
- D)

SECTION-II

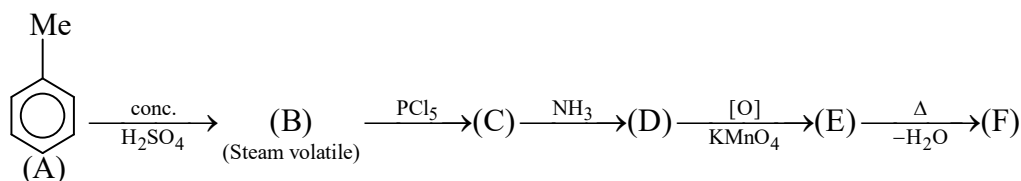
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only**. Have to Answer any 5 only out of 10 questions and question will be evaluated according to the following marking scheme:

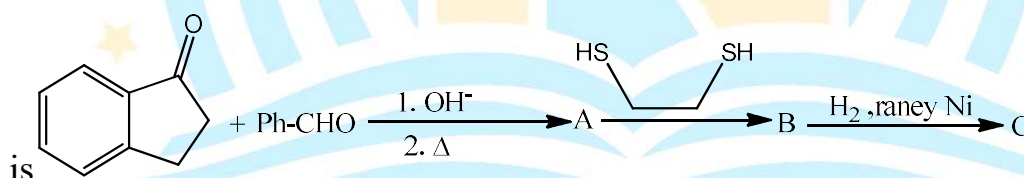
Marking scheme: +4 for correct answer, -1 in all other cases.

51. A catalyst lowers activation energy of a reaction from 4.2 kJ mol^{-1} to 'X' kJ mole^{-1} . The rate of reaction in the absence of the catalyst at 50 K is equal to the rate of the reaction in the presence of catalyst at 47.6 K. The value of 'X' is (nearest integer):
52. How many of the following are stronger than acetic acid?
 ClCH_2COOH , NCCH_2COOH , $\text{CH}_3\text{CH}_2\text{COOH}$, HClO_4 , HCl , $\text{C}_6\text{H}_5\text{OH}$, $\text{C}_2\text{H}_5\text{OH}$
53. The weight in grams of potassium dichromate required to produce 5.6 L of CO_2 at STP (1 bar, 273 K) from its reaction with excess of oxalic acid in acidic medium is X grams. The value of $\frac{X}{3}$ is _____ (K=39, Cr=52, O=60)

54. The total number of moles of Grignard reagent (CH_3MgBr) acting as a base require to react with the final major product F is



55. A pale green crystalline metal salt of M dissolves freely in water. On standing it gives a brown ppt on addition of aqueous NaOH. The metal salt solution also gives a black ppt on bubbling H_2S in basic medium. An aqueous solution of the metal salt decolourizes the pink colour of the permanganate solution. The oxidation state of metal in the black precipitate is:
56. How many of the following species have bond order equal to 1.33
 CO_3^{2-} , NO_3^- , BF_3 , O_3 , SO_3^{2-} , ClO_3^- , MnO_4^{2-} , CO , Graphite, benzene
57. Pure benzene freezes at 5.45°C . A solution containing 7.24g of $\text{C}_2\text{H}_2\text{Cl}_4$ in 115.3g of benzene was observed to freeze at 3.55°C . What is the molal freezing point constant of benzene (in K kg mole^{-1})?
58. Degree of unsaturation in the final major hydrocarbon product 'C'



59. The charge required in faradays for the oxidation of 1 mole of Mn_3O_4 into MnO_4^{2-} in the presence of alkali is _
60. Treating 12.8 g of calcium carbide with water produces compound 'A' with 50% yield. 'A' on passing through red hot iron tube at 873 K gives compound B with 100% yield. Compound B upon sulphonation and then treating with NaOH produces C with 80% yield. Find the mass of salt C (in gm) formed (nearest integer)

MATHEMATICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. If Z_1, Z_2 and Z_3, Z_4 are two pair of conjugate complex numbers then $\arg\left(\frac{Z_1}{Z_4}\right) + \arg\left(\frac{Z_2}{Z_3}\right)$ equals to
- A) $\frac{\pi}{2}$ B) $-\frac{\pi}{2}$ C) 0 D) π
62. The system of equations
- $$\alpha x - y - z = \alpha - 1$$
- $$x - \alpha y - z = \alpha - 1$$
- $$x - y - \alpha z = \alpha - 1$$
- has no solution, if α is
- A) either -2 or 1 B) -2 C) 1 D) 3
63. If $x^2 + 3x + 5$ is the greatest common divisor of $(x^3 + ax^2 + bx + 1)$ and $(2x^3 + 7x^2 + 13x + 5)$ then $10(a+b)$ is
- A) 80 B) 84 C) 88 D) 90
64. **Statement –1:** If $f: \{1, 2, 3, 4, 5\} \rightarrow \{1, 2, 3, 4, 5\}$ then the number of onto functions such that $f(i) \neq i$ is 42
- Statement –2:** If n things are arranged in row, the number of ways in which they can be de-arranged so that no one of them occupies its original place is
- $$n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \frac{1}{n!} \right)$$
- Statement–3:** Number of ways of distribution of 12 identical balls into 3 identical boxes is 19
- Statement –4:** Number of ways of distribution of n identical objects among r persons, each one of whom can receive any number of objects is ${}^{n+r-1}C_{r-1}$
- Which of the following must be truth value of above statements in that order.
- A) FTFT B) FFTF C) FTTT D) TFTF

65. $A = \{x / x \text{ is matrix containing 32 identical elements}\}$, $B = \{y / y \text{ is matrix containing 56 identical elements}\}$, A matrix is chosen at random from each of the sets A and B then the probability that $(x \times y)$ is possible is

- A) $\frac{1}{9}$ B) $\frac{1}{11}$ C) $\frac{1}{12}$ D) $\frac{1}{14}$

66. Each of the persons A and B independently tosses three fair coins. The probability that both of them get the same number of heads is

- A) $\frac{5}{8}$ B) $\frac{5}{16}$ C) $\frac{1}{2}$ D) $\frac{1}{8}$

67. The probability distribution of random variable "X" is given by

X	1	2	3	4	5
P(X)	K	2K	2K	3K	k

then $P(1 < X < 4 / X < 3)$ is

- A) $\frac{1}{3}$ B) $\frac{2}{3}$ C) $\frac{1}{2}$ D) $\frac{1}{4}$

68. Statement 1: $2^{4n} - 2^n (7n+1)$ is divisible by square of 14 $\forall n \in N$

Statement 2: $(1+x)^n = 1 + {}^nC_1 x + \dots + {}^nC_n x^n \forall n \in N$

A) Both Statement - 1 and Statement - 2 are true and Statement - 2 is the correct explanation of Statement - 1

B) Both Statement - 1 and Statement - 2 are true but Statement - 2 is not the correct explanation of Statement - 1

C) Statement - 1 is true, Statement - 2 is false

D) Statement - 1 is false, Statement - 2 is true

69. If $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$, where $0 < x \leq 1$ then

A) Both $f(x)$ and $g(x)$ are increasing functions

B) Both $f(x)$ and $g(x)$ are decreasing functions

C) $f(x)$ is increasing and $g(x)$ is decreasing

D) $g(x)$ is increasing and $f(x)$ is decreasing

70. Let $\vec{a} = 4\vec{i} + 3\vec{j}$ and $\vec{b} = 3\vec{i} - 4\vec{j} + 5\vec{k}$. If $\vec{c}$ is a vector such that $\vec{c} \cdot (\vec{a} \times \vec{b}) + 25 = 0$, $\vec{c} \cdot (\vec{i} + \vec{j} + \vec{k}) = 4$, and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals

- A) $\frac{1}{5}$ B) $\frac{3}{\sqrt{2}}$ C) $\frac{1}{\sqrt{2}}$ D) $\frac{5}{\sqrt{2}}$

71.

Column-I	Column-II
A) $\sin\left(2\tan^{-1}\frac{3}{4}\right)$	P) $\frac{14}{15}$
B) $\cos\left(2\tan^{-1}\frac{1}{7}\right)$	Q) $\frac{3}{5}$
C) $\sin\left(4\tan^{-1}\frac{1}{3}\right)$	R) $\frac{2}{\sqrt{13}}$
D) $\cos 2\left(\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{2}{9}\right)$	S) 1
	T) $\frac{24}{25}$

A) A-T; B-T; C-T; D-Q

B) A-R; B-T; C-S; D-Q

C) A-Q; B-T; C-R; D-R

D) A-T; B-S; C-S; D-T

72. Let the tangent to the circle $C_1 : x^2 + y^2 = 2$ at the point $M(-1,1)$ intersect the circle $C_2 : (x-3)^2 + (y-2)^2 = 5$, at two distinct points A and B. If the tangents to C_2 at the points A and B intersect at N, then the area of the triangle ANB is equal to

- A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{1}{6}$ D) $\frac{5}{3}$

73. The equation of the curve obtained by reflecting the parabola $y^2 = 4x$ about the line $x - y + 13 = 0$ is

A) $(2y - x - 13)^2 = 4(y + 13)$

B) $(2y + x - 13)^2 = 4(y - 13)$

C) $(2y - x - 13)^2 = 4(y - 13)$

D) $(x + 13)^2 = 4(y - 13)$

74. The line $y = x + 1$ meets the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ at two points P and Q. If r is the radius of the circle with $\overline{PQ}$ as diameter, then $(3r)^2$ is

A) 20

B) 12

C) 11

D) 8

75. If $\int e^{\sin x} (x \cos^3 x - \sin x) \sec^2 x \, dx$ is $e^{\sin x} f(x) + C$ (C is a constant). Then $f(0)$ is
- A) 0 B) -1 C) 1 D) $\frac{\pi}{2}$
76. The value of $\int_0^{2\pi} [\sin 2x(1 + \cos 3x)] dx$, where $[t]$ denotes the greatest integer function, is
- A) 2π B) -2π C) $-\pi$ D) π
77. The area bounded by the curve $|px + qy| + |qx - py| = a$, is _____ squnits.
- A) $2a^2$ B) $2a^2 \sqrt{p^2 + q^2}$ C) $\frac{2a^2}{\sqrt{p^2 + q^2}}$ D) $\frac{2a^2}{p^2 + q^2}$
78. STATEMENT-1: Number of circles passing through the points $(1, 2), (3, \frac{1}{2}), (\frac{1}{3}, \frac{5}{2})$ is one.
STATEMENT-2: Through three non-collinear points in a plane only one circle can be drawn.
- A) Both Statement - 1 and Statement - 2 are true and Statement - 2 is the correct explanation of Statement - 1
B) Both Statement - 1 and Statement - 2 are true but Statement - 2 is not the correct explanation of Statement - 1
C) Statement - 1 is true, Statement - 2 is false
D) Statement - 1 is false, Statement - 2 is true
79. If $|r| > 1$ and $x = a + \frac{a}{r} + \frac{a}{r^2} + \dots$, $y = b - \frac{b}{r} + \frac{b}{r^2} - \dots$, $z = c + \frac{c}{r^2} + \frac{c}{r^4} + \dots$, then $\frac{xy}{ab} =$
- A) $\frac{c}{z}$ B) $\frac{z}{c}$ C) cz D) $\frac{1}{cz}$
80. Let $X = \{x \in N : 1 \leq x \leq 17\}$ and $Y = \{ax + b : x \in X \text{ and } a, b \in R, a > 0\}$
If mean and variance of element of Y are 17 and 216 respectively then $a+b$ is
- A) -7 B) 7 C) 9 D) -27

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. If Z_1, Z_2, Z_3 are the three points lying on the circle $|Z| = \sqrt{3}$, then maximum value of $|Z_1 - Z_2|^2 + |Z_2 - Z_3|^2 + |Z_3 - Z_1|^2$ is

82. Let A be a non-singular matrix such that $A^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 6 & 8 \\ 100 & 100 & 99 \end{pmatrix}$. Then the value of $\det(\text{adj}(2A)) + \det(2A) =$
83. Let $f(x) = x^2 - 4x - 3, x > 2$ and $g(x)$ be the inverse of $f(x)$. Then the value of $\frac{1}{g'(2)}$, is (here g' represents the first derivative of g)
84. The sum of all the local extreme values of the twice differentiable function $f: R \rightarrow R$ defined by $f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1)$ is λ , then $\lambda + 30$ is
85. Let $C(\alpha, \beta)$ be the circumcentre of the triangle formed by the lines $4x + 3y = 69$, $4y - 3x = 17$, and $x + 7y = 61$. Then $(\alpha - \beta)^2 + \alpha + \beta$ is equal to
86. If $f(x+y) = f(x) + f(y) + 2xy - 1, \forall x, y \in R$ and f is a differentiable function and $f'(0) = 3$, then number of non-differentiable points for $y = \|f(x) - 1\| - 1$ is/are
87. Area bounded by the curves $y = x^2$ and $y = \frac{2}{1+x^2}$ is $\pi - \lambda$ then 3λ is
88. 5 married couples have to be seated around a circular table. Let P_1 be the probability that every man is seated adjacent to his wife and P_2 be the probability that every man is seated diametrically opposite to his wife's then value of $\frac{P_1}{P_2}$ is
89. If $\lim_{x \rightarrow 0} \frac{ae^x + b \cos x + c + dx}{x \sin^2 x} = 3$, then the value of $a + 2b + c + d$ is equal to
90. Let $f: R - \{3\} \rightarrow R - \{1\}$ be defined $f(x) = \frac{x-2}{x-3}$
 $g: R \rightarrow R$ be given as $g(x) = 2x - 3$ then the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to